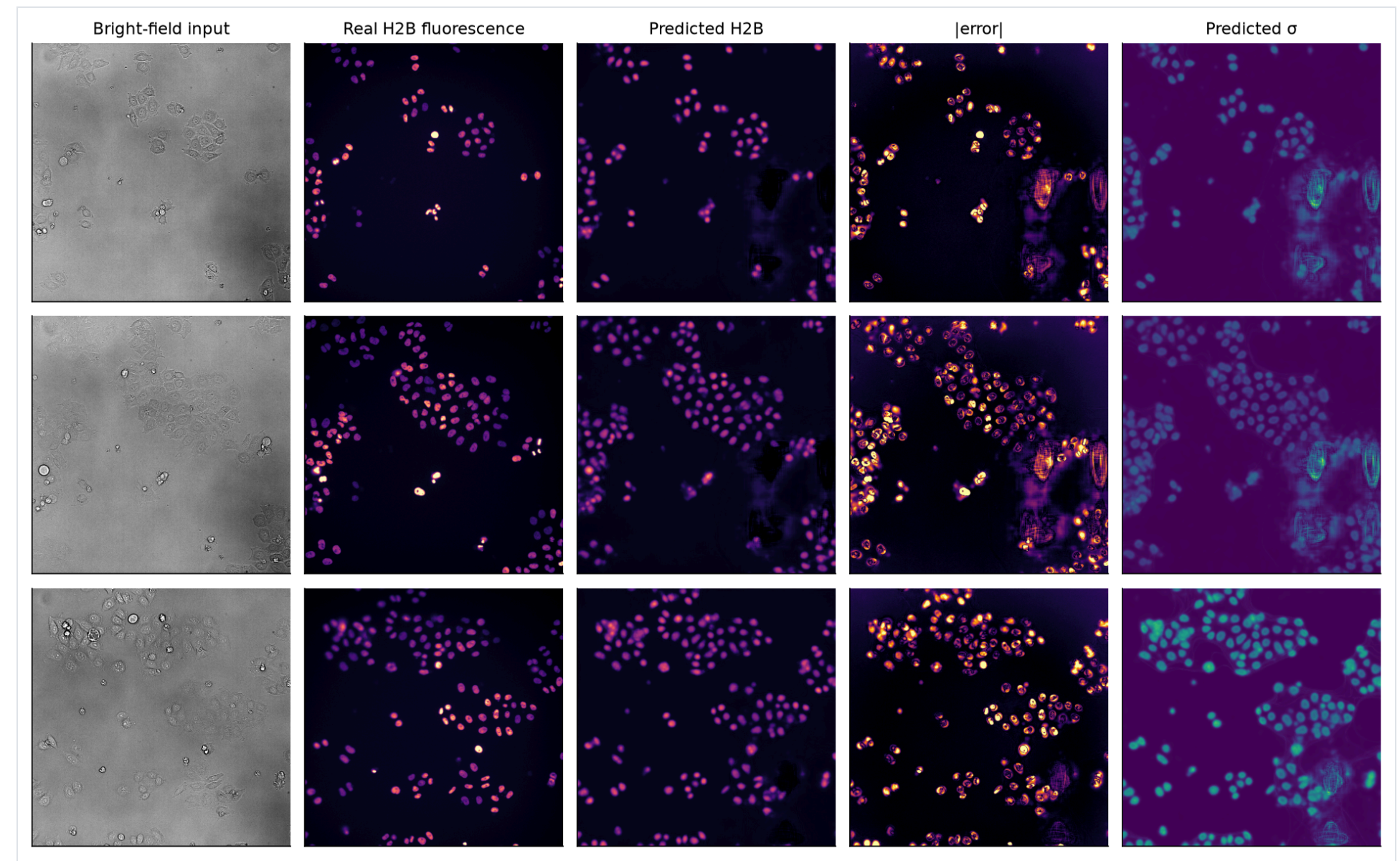


ChipStain

Label-free nuclear staining that tells you where not to trust it



- Team Hack2Publish — Md. Hamid Hosen, Esfer Sami, Foysal (CSE students)

Nuclear read-outs need a stain — and a stain has a cost

- Cell number, growth and viability on an organ-on-a-chip all start from a nuclear stain
- Fixation ends the experiment; live DNA dyes are phototoxic
- Reporter lines are rarely available for iPSC-derived tissue such as neurons
- In-silico labeling predicts the stain — but a point prediction gives no warning when it is wrong

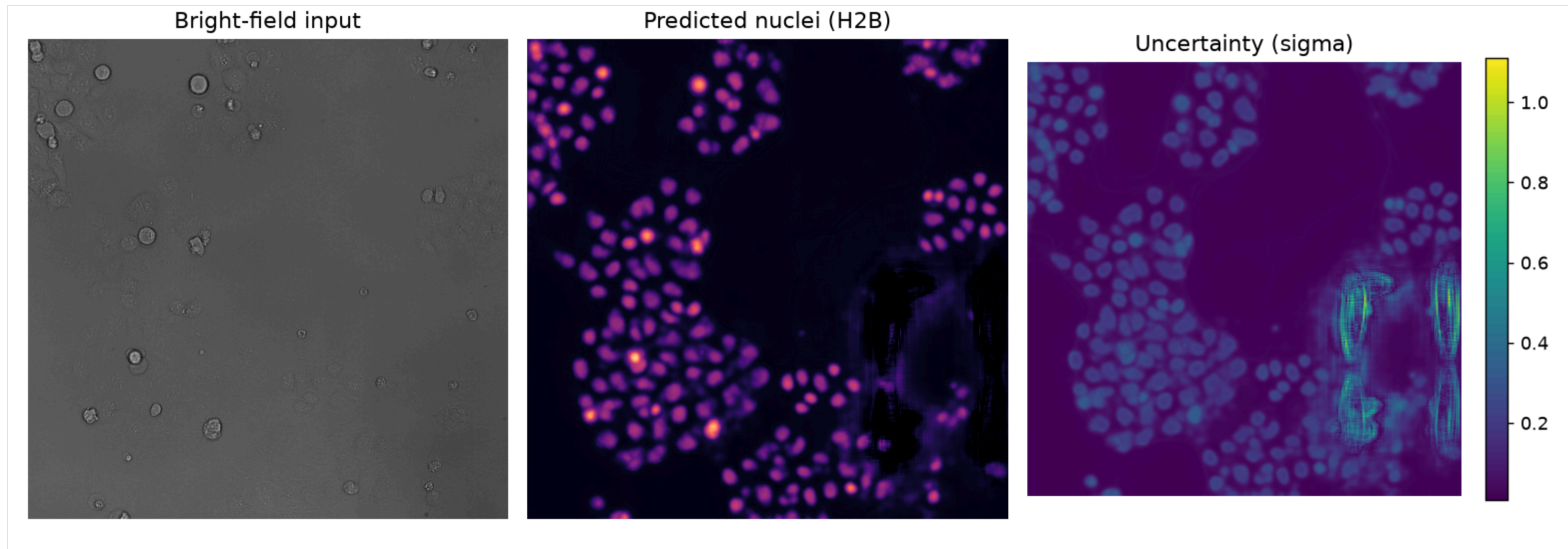
Predict the stain and its uncertainty



- U-Net, ImageNet ResNet-34 encoder; head predicts mean μ and variance σ^2 per pixel
- β -NLL loss keeps the variance from swallowing hard pixels
- 8× test-time augmentation: $\sigma^2 = \text{learned variance} + \text{TTA disagreement}$
- No new mechanism — the contribution is testing whether σ actually helps

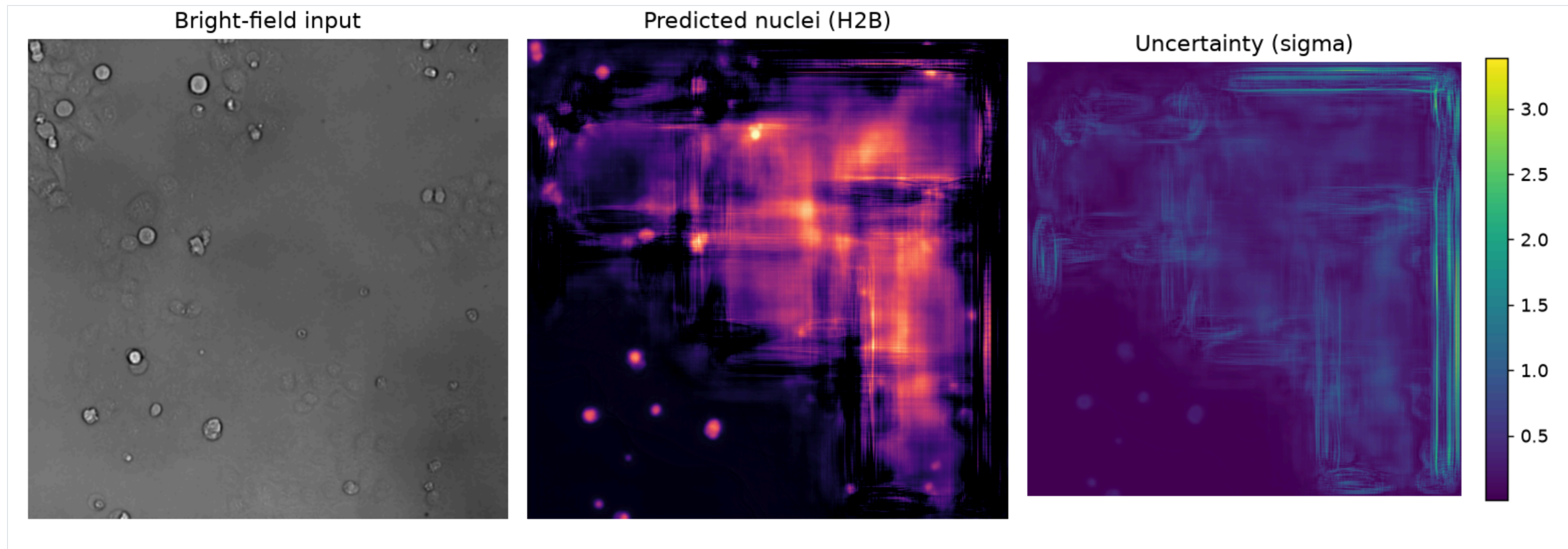
183 nuclei found — the reference has 180

- Bright-field in → predicted H2B and σ out, on a laptop CPU
- σ is higher on nuclei and near zero on background
- Mean $\sigma = 0.08$



The prediction fails — and σ says so

- Mean σ jumps from 0.08 to 0.33
- The app warns: input looks unlike the training data
- A plain U-Net becomes less uncertain here — it fails silently



Same fidelity, much better uncertainty

Pearson r

0.768 vs 0.764

$\rho(\sigma, \text{error})$

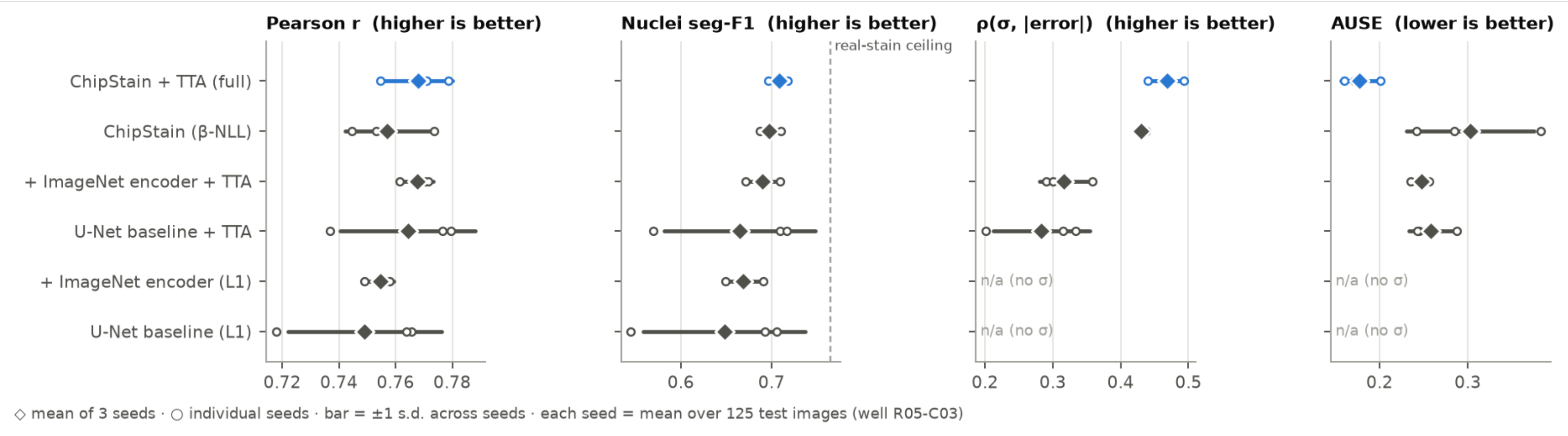
0.47 vs 0.28

95 % interval coverage

92 %

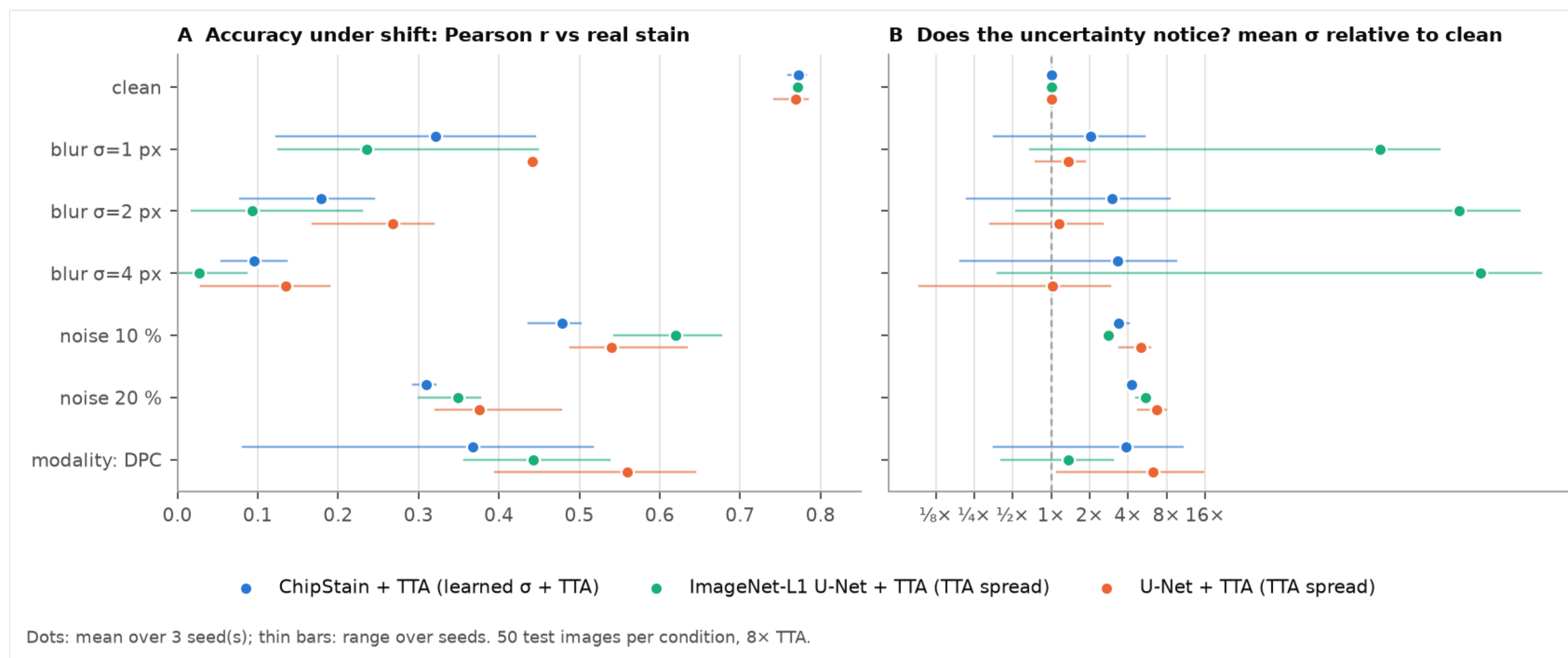
MAE cost

+10 %



When the input drifts, σ rises — the baseline's falls

- Blur and phase-contrast input break both models' predictions
- ChipStain's mean σ rises several-fold and flags the failed images
- The U-Net's TTA spread falls under blur



60 hours of growth from bright-field alone

doubling time, label-free

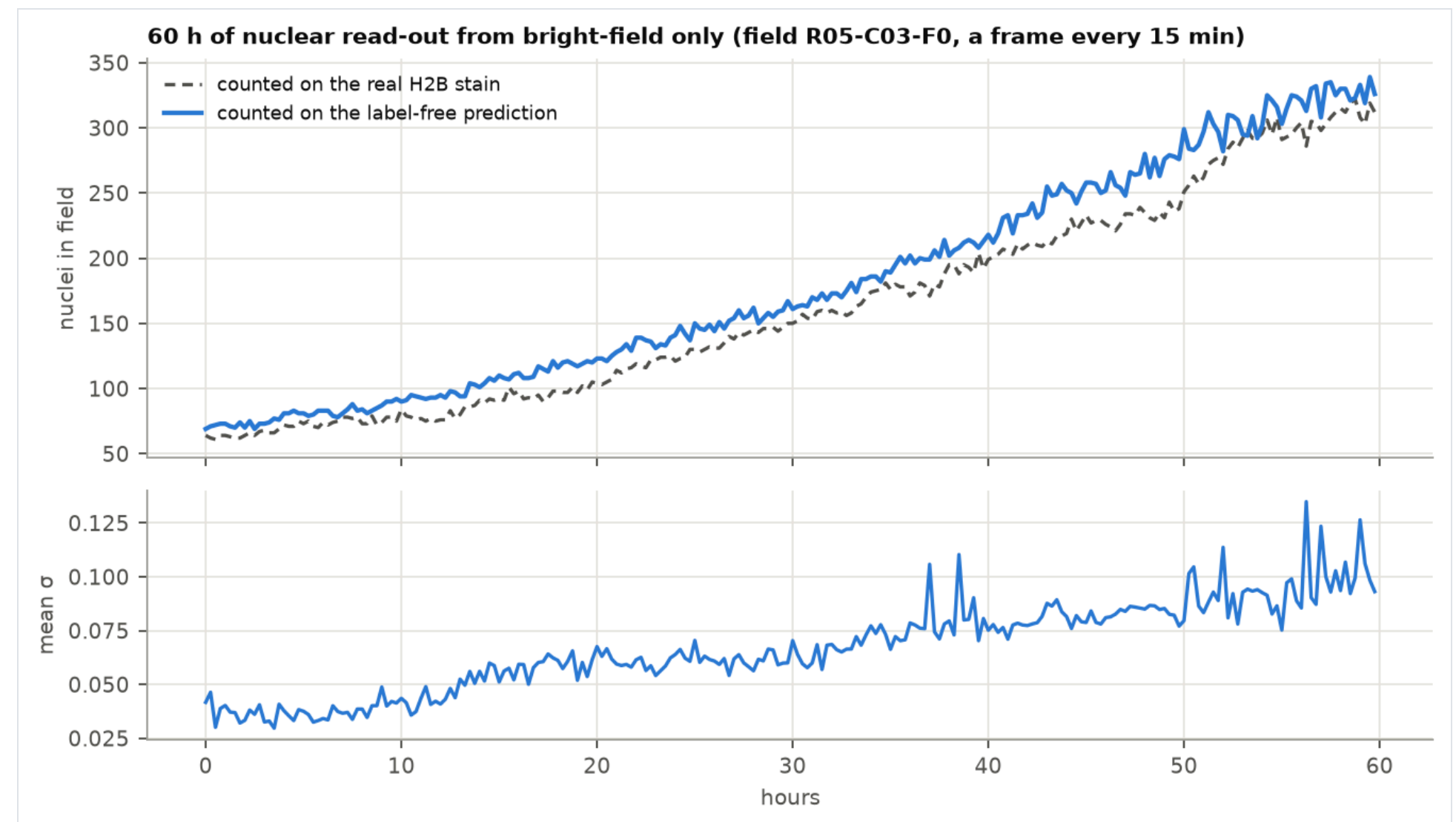
24.9 h

from the real stain

23.9 h

error without $\rightarrow$ with σ gate

9.1 $\rightarrow$ 4.1 %



What σ does not do

- For counting alone, direct segmentation wins: Cellpose F1 0.831 vs ours 0.708
- Per-nucleus σ is no better than a simple brightness rule
- No organ-on-chip images were available — validation is in a plate and a dish
- Neurons (iPSC-derived, in a dish): zero-shot r 0.59 $\rightarrow$ 0.76 after fine-tuning on one well

github.com/hamidhosen42/AI4S-Open-Innovation-AI-for-Life-Science

- Code, seed-0 weights, data links, CPU inference script
- One script regenerates every number, figure, this deck and the report
- Team Hack2Publish — Md. Hamid Hosen · Esfer Sami · Foysal (CSE students)